Essay in ML/DL in Physics

Goal

The essay should focus on an application of Machine Learning and/or Deep Learning in a specific area of physics of your choice. It must address a well-defined problem and describe the methodology used to solve it. Beyond presenting the objectives and the achieved results, the work should clearly highlight the architecture of the ML model and explain how it operates.

Choosing the subject

You can choose any theme that is inside of the goal. If necessary, for inspiration, you can see these references:

- https://www.frontiersin.org/journals/physics/articles/10.3389/fphy.2024.1322162/full
  - Chapter 3 is organized with examples of different areas of physics
- https://www.sciencedirect.com/science/article/pii/S0957417424015458?via%3Dihub
  - Focused more on the architecture of the physics informed ML (PIML)
- https://link.springer.com/article/10.1007/s44379-025-00016-0
  - Review of physics informed ML (PIML)

The choice of the subject should be made until: 04/05/2026

Format

Oral presentation

For the presentation, a visual support tool such as PowerPoint must be provided. The presentation may not exceed 30 minutes and will be followed by a discussion.

The days of the presentations are:

Name	Nº	Date
Julia Zofia Jurasz	76173	25/05/2026
Maria João Ferreira Guedes Morais	73957	18/05/2026
Maria Mafalda Raposo Ribeiro	69686	11/05/2026
Naiana Pedro Lima	73395	25/06/2026
Ricardo Beja Segundo	73872	11/05/2026
Sebastião Borges Parreira Mendes Gomes	75094	18/05/2026
Sofia Alonso Durão Lopes	75671	25/05/2026

Support Document

A support document must also be provided in one of two formats:

- A written document in which all information is presented. It should be between 10 and 15 pages in length.
- A visual presentation, where each slide is accompanied by a one-page support text explaining the content intended for that slide.

The deadline for submitting the document is: 01/06/2026.